Assignment : 06

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Here , my student login EUID is : gg0640

So , n+1 = 0+1 = Chi-Square Test if n+1=1, 5, 9

Step 1 : Imported the required libraries .

Here I had imported stats library present in scipy module

For visualization , we had imported matplotlib.pyplot which is built matplotblib.

Step 02:

Here " variable = np.linspace(0 , 20,100) "

This function linspace() from numpy , this enables us in creating arrays matrices and some functions which are related to mathematics.

a. Linspace is used to create an array of even spaced numbers or values of a specified interval.
b. 0,20,100 : this condition in linspace creates us values from 0 to 20 which 100 points in between them .
c. This allows you to have some range of values to work with .

Step 03:

degrees_of_freedom = [1, 4, 6, 7]

Here , these are the values here that are allowed to vary very freely . so this each degree of freedom value corresponds us to a specific scenario in our tests .

So , if we are having a table with 2 rows and 2 tables dataset , the degrees of freedom would be

Degrees of freedom = (rows num -1) * (columns number -1)

Which is (2-1)*(2-1) =1 .

Which results in a specific scenario .

Step 04 :

plt.figure(figsize=(10, 6))

This is basically done to create a visualization plot in the output which the given size and it is done using matplotlib in python .

The syntax is figure(figsize=(width,height) which plots according to the user given ratios.

Step 05:

In step 05 we basically iterated over the degrees of freedom which is done using the for loop .

a. So by iterating it and then the plotting of corresponding chi square with the probability density is done . This basically means that that this allow us to visualize how the shape of the chi test is being altered with application to different degrees of freedom
b. So , here for each of the value for degrees of freedom , it calculates the pdf for the chi squared using the .pdf() function imported from the stats module and then it plots the pdf against the variable using plot .
c. Then the plt.plot(variable , chi_sq_pdf,label=f'df={df}') :The plotting is done here with the chi squared values are being plotted against pdf values , a label argument is also allowed in the pdf() function which indicates the degrees of freedom as a label on the plot .

Step 06 :

Plotting :

Then the plotted graph is further enhanced. Here , the title and the x and y labels are being plotted .

Here , we can choose to display the legend , by using plt.legend() we can display the legend on the plot which takes the labels from the for loop including the degrees of freedom .

We can also choose to have the grid in the plot or not . This can be done by using the plt.grid(False) this takes Boolean value true or false to display the grid or not .

Plt.show() : this element is used to display the plot which is configured above and this is the must step to display the plot .

Code

my student login id is gg0640

so n= 0

0+1= 1 : chi-Squared test

from scipy import stats # importing libraries for stats and maths and visualization import numpy as np

import matplotlib.pyplot as plt

variable = np.linspace(0, 20, 100) # Generating values from 0 to 20 with 100 underoying points

degrees_of_freedom = [1, 4, 6, 7] # Different degrees of freedom ranging from 1,7

print(variable)

plt.figure(figsize=(10, 6)) # Setting figure size of the ploting size

for df in degrees_of_freedom:# iterating over a for loop with deg of freedom

    chi_sq_pdf = stats.chi2.pdf(variable, df)   # pdf for chi squared using pdf function

    plt.plot(variable, chi_sq_pdf, label=f'df={df}') # plotting

plt.title('Chi-Square test')

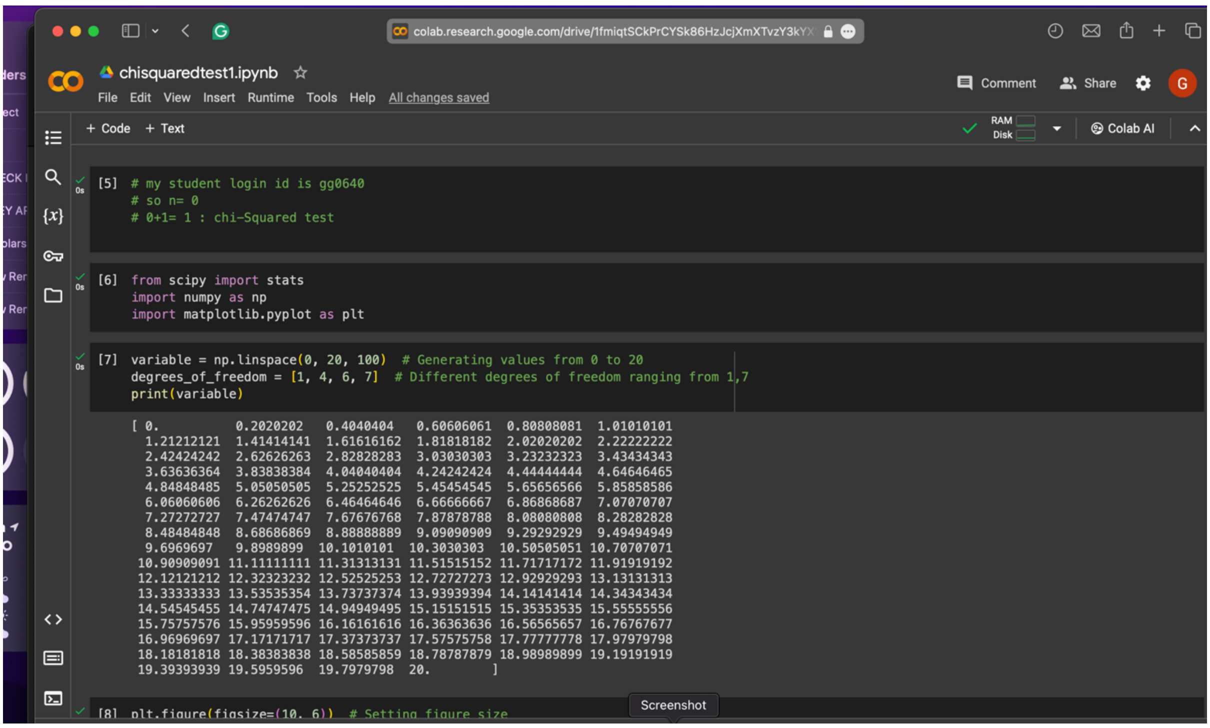
plt.xlabel('Values on x axis ')

plt.ylabel('Density')

plt.legend()

plt.grid(False)

plt.show()



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[8] plt.figure(figsize=(10, 6))  # Setting figure size

<Figure size 1000x600 with 0 Axes>
<Figure size 1000x600 with 0 Axes>
```

```
for df in degrees_of_freedom:
    chi_sq_pdf = stats.chi2.pdf(variable, df)  # Probability density function for chi-square
    plt.plot(variable, chi_sq_pdf, label=f'df={df}')  #Plotting chi-square PDF for each degree of freedom
plt.title('Chi-Square test')
plt.xlabel('Values on x axis ')
plt.ylabel('Density')
plt.legend()
plt.grid(False)
plt.show()
```